Highlights

FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning

- FOLoRA protects the Fisher-important directions of a shared LoRA adapter.
- Protected directions are weighted by Fisher importance, not protected equally.
- FOLoRA minimizes the leading term of a second-order bound with explicit remainder.
- Fixed trainable-parameter budget with no replay and no inference overhead.
- One setting transfers across benchmarks; EWC-LoRA's default strength does not.

FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning

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Abstract

Parameter-efficient fine-tuning (PEFT) methods such as low-rank adaptation (LoRA) freeze a pre-trained backbone and train few parameters, suiting continual learning on resource-constrained devices. However, sequentially fine-tuning a shared LoRA adapter still forgets catastrophically: the low-rank subspace is overwritten by new tasks. We propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method protecting the directions past tasks deem important. FOLoRA accumulates, per adapted layer, a second-order importance kernel from the empirical Fisher on the adapter parameters. Its top eigendirections form an *importance-weighted protected subspace*, and a regularizer penalizes alignment of the adapter displacement with them, weighted by eigenvalue. This yields a stability–plasticity trade-off between preserved and reusable directions. We further derive a second-order forgetting bound with an explicit remainder and show FOLoRA's regularizer minimizes its leading term; we audit that bound's slack empirically, using it as a design principle rather

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than a numerical predictor. Under the frozen-feature nearest-class-mean protocol, FOLoRA matches the strongest regularized baseline on Split CIFAR-100 and Split ImageNet-R; one setting is used unchanged on both, where EWC-LoRA's pre-registered default strength does not transfer. Unlike per-task-adapter methods (O-LoRA, InfLoRA), FOLoRA needs no replay and no new module per task ($\sim 18\times$ fewer trainable parameters), adds no inference cost, and merges into the backbone; only the classifier head grows with the task count, shared by all methods and discarded at evaluation under this protocol. We also quantify the trade-off: a rank-k importance model rather than EWC's diagonal costs a larger footprint.

Keywords: continual learning, catastrophic forgetting, parameter-efficient fine-tuning, low-rank adaptation, Fisher information

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1. Introduction

Continual learning aims to learn a sequence of tasks without catastrophically forgetting previously acquired knowledge [1]. The dominant strategies — rehearsal [2], regularization [3, 4], and dynamic architectures [5] — were largely developed in the regime of training a model from scratch. Recently, however, the community has shifted toward *pre-trained models* (e.g., Vision Transformers) combined with *parameter-efficient fine-tuning* (PEFT), in which the backbone is frozen and only a tiny set of parameters is learned per task [6, 7, 8]. This shift has produced a fast-moving literature on prompt-, adapter-, and subspace-based PEFT for continual learning [9, 10, 11, 12, 13].

Low-rank adaptation (LoRA) [14] is a particularly attractive PEFT primitive for continual learning, and recent variants exploit it in different ways [15,

16, 17, 18]: a shared LoRA adapter keeps *its own* trainable parameter count fixed regardless of the number of tasks, which is ideal for resource-constrained deployment. However, sequentially fine-tuning a shared adapter still suffers from catastrophic forgetting — although the number of trainable parameters is small, the low-rank subspace is repeatedly overwritten, and the representational structure learned for earlier tasks is destroyed.

A line of recent work addresses this by allocating *orthogonal* low-rank subspaces to different tasks [8, 19], and refined task-wise allocation schemes continue to appear [20, 21, 22]. These methods prevent interference by making the new task's adapter (nearly) orthogonal to those of previous tasks. While effective, they exhibit two limitations. First, they protect all directions of the past subspace *equally*, ignoring the fact that different directions carry vastly different amounts of information about a task. Second, most of them allocate a *new* adapter per task, so the parameter count grows linearly with the number of tasks.

In this paper we propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method that mitigates forgetting by protecting past tasks' *important* directions in proportion to their importance. Our contributions are:

1. We frame forgetting in low-rank continual learning through a second-order lens and derive an upper bound that decomposes forgetting into Fisher-weighted subspace overlaps. We audit the slack of that bound empirically rather than assuming it away, and use it as a design principle for what to penalize rather than as a numerical predictor of how much forgetting a given setting removes (Sec. 4).

2. We propose FOLoRA, which estimates a per-layer second-order importance kernel from the empirical Fisher information restricted to the adapter parameters, forms an importance-weighted protected subspace from its top eigendirections, and regularizes the adapter to stay orthogonal to these directions (Sec. 3).
3. We show that FOLoRA's regularizer is exactly the Lagrangian penalty for constraining the leading term of a second-order forgetting bound, realizing a principled stability–plasticity trade-off, and that it contains equal-weight orthogonalization — the geometry underlying O-LoRA-style subspace methods — as the special case in which all protected directions carry equal weight.
4. On Split CIFAR-100 and Split ImageNet-R, FOLoRA matches the strongest regularized baseline in average accuracy with no replay buffer and no growth in parameters, while improving significantly over the parameter-efficient prompt and subspace baselines. Its main configuration is selected once and carried across both benchmarks unchanged, whereas EWC-LoRA's pre-registered default strength does not transfer: at λ=100 it sits 2.8 points below EWC-LoRA's own tuned optimum on ImageNet-R and is the weakest point of its grid on CIFAR-100. We also quantify the importance-model memory FOLoRA pays for its rank-k subspace representation relative to EWC's diagonal one, and we report the run-to-run reproducibility floor that sets the resolution of these comparisons (Sec. 5).

2. Related Work

Continual learning. Classical continual learning methods fall into three families. Rehearsal methods store and replay a subset of past data [2]; regularization methods add a penalty that slows down learning on parameters important to past tasks, most notably EWC via the diagonal Fisher information [3] and SI/MAS via synaptic importance [4, 23]; dynamic methods grow the network per task [5, 24]. Recent surveys of the field appear in [13, 12, 25], and the exemplar-free, representation-level, and distillation-based variants most relevant here in [26, 27, 28, 29]. Our method is a regularization method, but it operates on the low-rank subspace of a frozen backbone rather than on raw parameters.

Parameter-efficient continual learning with pre-trained models. A recent and effective line of work keeps a pre-trained backbone frozen and learns only prompts or adapters; the space is surveyed in [9, 10, 11, 30]. Prompt-based methods — L2P [6], DualPrompt [31], and CODA-Prompt [7] — learn a pool of prompts and select or compose them per input. LoRA-based methods attach low-rank adapters. Related designs differ in where they spend the per-task budget: expandable or branched subspaces [20, 16, 17, 21], decoupled scaling rules [15], orthogonal adapter banks [32], further adapter variants [33, 34, 35, 36], dual-consolidation and hierarchical prompt decompositions [37, 38], and prototype- or prompt-routing schemes [39, 40, 41, 42]; for the online and task-free regime, see [18].

Orthogonalization and subspace allocation. A distinct line makes *where* a new task may write the design variable, rather than how strongly. Its classical

form is gradient projection: orthogonal gradient descent [43], GPM [44] and the null-space-of-feature-covariance update of Adam-NSCL [45] retain a basis describing past tasks and constrain new updates to its orthogonal complement, a construction that also admits generalization guarantees [46]. The same idea reappears in PEFT as subspace allocation for low-rank adapters: O-LoRA [8] orthogonally initializes each task's adapter against previous tasks' input subspaces, and InfLoRA [19] places the new task's subspace in the intersection of the new task's gradient space and the orthogonal complement of the old tasks' gradient space, removing interference while retaining plasticity. The same axis is now being explored for prompt tuning, via null-space constructions [47].

Two properties are shared across this line, and they are what FOLoRA changes. First, the protected subspace is obtained from a *structural* criterion — the past tasks' input or feature covariance, a gradient subspace, or an explicit orthogonality constraint — rather than from the curvature of the past loss. Second, within the protected subspace every retained direction is defended equally: the budget decides *how many* directions to keep, never *how much* each one matters. FOLoRA keeps the subspace framing and replaces both choices with a single object, the adapter block of the accumulated empirical Fisher, so that the retained directions are those in which the past loss is most curved and each is penalized in proportion to its curvature. The equal-weight geometry of this family is then the special case $\sigma_j \equiv 1$ of FOLoRA's penalty (Sec. 4), and we compare against O-LoRA and InfLoRA empirically in Sec. 5.2.

Second-order methods and curvature-weighted penalties. EWC [3] uses the diagonal Fisher as a per-parameter importance weight. FOLoRA combines that idea with the subspace framing above: it uses the top-k spectrum of the adapter block of the second-order expansion — not the diagonal Fisher, and not the full $d \times d$ matrix either — and applies it as a soft, importance-weighted orthogonal projection rather than a hard null-space constraint, which avoids the rank saturation that hard projections suffer once the null space shrinks. Recent analyses of flatness-aware optimization approach the same goal from other angles [48], flatness itself has been connected to generalization bounds [49], and gradient-norm regularization acts on the same stability–plasticity axis without a curvature weighting [50].

Fisher information inside low-rank adaptation. The two ingredients FOLoRA combines — empirical Fisher information and low-rank adaptation — have met before, and the differences are worth stating precisely rather than leaving to the reader. FR-LoRA [51] is the closest in problem setting: it targets continual learning and regularizes LoRA adapters with Fisher information. It differs in *what* is regularized — a per-task adapter, so the trainable budget grows with the sequence exactly as in the branch-per-task family above — and in the role the Fisher plays, a parameter-importance penalty in the spirit of EWC rather than a spectral subspace of the adapter displacement. Three further works use Fisher information for LoRA but not for continual learning: FiLoRA [52] uses it to guide the low-rank update, FI-LoRA [53] to decide how much rank each layer receives dynamically, and the Fisher-subspace construction of Feng et al. [54] to choose an initialization rather than to constrain training — an initialization changes where training

begins, whereas FOLoRA's penalty acts at every step. None of the three faces a task sequence, and none of the four penalizes the alignment of a *shared* adapter with an importance-weighted past subspace, which is the object this paper derives and evaluates. The scope of that negative claim is a search for work combining Fisher information with low-rank adaptation, as of this writing; the four works are listed together here so that the novelty claim can be checked rather than taken on trust.

Theory of forgetting. Second-order analyses of forgetting are well established for models trained from scratch: the forgetting of a task is governed by the curvature of its loss landscape and by the overlap between successive tasks' input distributions or NTKs [55, 56, 57, 58]. We build on this line but instantiate it in the low-rank adapter subspace of a frozen pre-trained model, and we use it constructively: the leading term of the bound is exactly the quantity that a subspace regularizer can minimize, which yields the FOLoRA objective (Sec. 4). Recent theory sharpens this picture for lazy-training and kernel regimes [59, 60], and the empirical counterpart — that low-rank adaptation forgets in a measurably different way from full fine-tuning [61] — is part of what motivates working in the adapter subspace rather than in the full parameter space.

Low-rank adaptation. LoRA [14] approximates weight updates by low-rank matrices and has become a standard PEFT primitive. Subsequent work studies initialization, rank allocation, and merging [36, 34]. We build directly on LoRA and extend it with a second-order anti-forgetting mechanism specialized for continual learning.

Evaluation protocols in PEFT-based continual learning. Results in this setting are strongly protocol-dependent. Keeping the standard trainable classification head that grows with the task sequence, class-incremental accuracy is typically in the single digits even when task-incremental accuracy is above 90%, because the loss is dominated by the classifier rather than by the adapted features; replacing the head at evaluation time with a nearest-class-mean classifier over frozen features raises the same checkpoints into the 70–80% range. Two consequences matter for how a method should be tuned and compared. First, hyperparameters — the regularization strength in particular — are only identifiable relative to the protocol under which they were selected, and an optimum found under a classifier-dominated metric need not be the optimum under the reported one. Second, a ranking established under one protocol does not transfer to another. Accordingly, all defaults reported in this paper are re-selected under the frozen-feature nearest-class-mean protocol, and every comparison we draw is made within that protocol. That the protocol can dominate the reported conclusion is now being recognized independently: recent work shows the class-incremental average is sensitive to evaluation choices and can misrepresent representation quality [62, 63, 64], that classifier bias and calibration distort comparisons unless handled explicitly [65, 66], and that simple frozen-feature classifiers over a pre-trained backbone remain competitive with more elaborate designs [67, 68, 69].

Gap addressed in this work. None of the lines above supplies what this paper sets out to provide. Subspace-based methods [8, 19, 45] protect a past task's subspace as a whole, so every retained direction is defended with the same strength regardless of how much the past loss actually depends on it,

and most of them allocate a branch per task, so the trainable parameter count grows with the sequence length. FR-LoRA is the one exception in setting but not in structure [51]: it inherits the same per-task allocation, so what it regularizes is a per-task adapter rather than a shared subspace, and its trainable count still grows with the sequence. Second-order analyses of forgetting [55, 58] characterize *why* forgetting happens but are developed for models trained from scratch and are not instantiated in a fixed low-rank adapter, nor do they yield an objective one can simply train with. What is missing is a regularizer that is *derived* from a forgetting bound carrying an explicit remainder, that weights protected directions by their curvature rather than uniformly, and that keeps the parameter budget constant across tasks. The derivation is used here as a statement of *which* quantity to penalize rather than as a numerical prediction of *how much* forgetting a setting removes: Sec. 4 audits the bound's slack empirically instead of assuming it away, and the method's empirical justification is the ablations in Sec. 5.3. Prior work in this journal addresses adjacent aspects — gradient-based task organization [22], gradient-norm regularization for dissimilar tasks [50], and task-adaptive routing [39] — but none of it derives the protected directions from a forgetting bound, which is the specific step taken here. FOLoRA is constructed to fill that gap.

3. Method

3.1. Preliminaries

Low-rank adaptation (LoRA). LoRA [14] freezes a pre-trained weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and injects a trainable low-rank residual,

$$W \;=\; W_0 \;+\; \Delta W \;=\; W_0 + \frac{\alpha}{r}\, BA, \tag{1}$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $r \ll \min(d_{\text{in}}, d_{\text{out}})$. The scalar α/r absorbs the rank normalization. The forward pass of an adapted layer is

$$h \;=\; W_0 x + \frac{\alpha}{r}\, BAx \;=\; W_0 x + \delta, \qquad \delta := \frac{\alpha}{r}\, BAx \in \mathbb{R}^{d_{\text{out}}}, \tag{2}$$

where δ denotes the adapter's residual contribution.

Class-incremental continual learning (CIL). We consider a sequence of T tasks $\{\mathcal{T}_t\}_{t=1}^{T}$. Task $\mathcal{T}_t$ provides a training set $\mathcal{D}_t = \{(x_i, y_i)\}$ over c new classes, and the classifier head is grown to cover all classes seen so far. At test time the task identity is not available, so prediction is made over the union of all seen classes. A single shared LoRA adapter is trained sequentially; its parameter count is $r(d_{\text{in}} + d_{\text{out}})$ per layer and is *independent* of T.

3.2. Motivation: forgetting as subspace overlap

Let ξ_τ collect all *trainable* parameters at the end of task $\mathcal{T}_\tau$ — the adapter parameters θ_τ (the flattened low-rank factors of Sec. 3.3) together with the classifier-head parameters — and let $\delta\xi = \xi - \xi_\tau$ be a later task's displacement of them. A second-order Taylor expansion of task τ's loss around ξ_τ gives

$$\mathcal{L}_\tau(\xi_\tau + \delta\xi) \;\approx\; \mathcal{L}_\tau(\xi_\tau) + \nabla_\xi \mathcal{L}_\tau^\top \delta\xi + \tfrac{1}{2}\, \delta\xi^\top \mathcal{F}_\tau\, \delta\xi, \tag{3}$$

where $\mathcal{F}_\tau = \mathbb{E}_{(x,y)\sim\mathcal{D}_\tau}\left[\nabla_\xi \log p(y|x)\, \nabla_\xi \log p(y|x)^\top\right]$ is the empirical Fisher information matrix of task τ over those parameters. At the end of task τ the parameters are (approximately) stationary for that task's loss, $\nabla_\xi \mathcal{L}_\tau(\xi_\tau) \approx 0$; this is stated as an explicit assumption (Assumption 1) and its residual under the regularizer is quantified in Sec. 4. Hence the *forgetting* of task τ — the increase in its loss caused by subsequent learning — is dominated by the quadratic term $\frac{1}{2}\,\delta\xi^\top \mathcal{F}_\tau\, \delta\xi$. Forgetting is therefore driven by the alignment of the new displacement with the directions in which task τ's loss is sharply curved (i.e., the top eigendirections of $\mathcal{F}_\tau$). Of that quadratic form, FOLoRA acts on the part involving the adapter parameters: its regularizer penalizes alignment of the adapter displacement $\delta\theta$ with the directions in which the adapter block of $\mathcal{F}_\tau$ is sharply curved. Sec. 4 makes this precise, including the head terms that FOLoRA does not constrain.

This observation separates FOLoRA from prior work: O-LoRA [8] treats every direction of the past low-rank subspace as equally important, whereas Eq. (3) shows that directions should be protected *in proportion to their curvature*. Protecting all directions equally over-constrains the adapter and wastes the fixed rank budget on directions that carry little information; weighting by curvature spends that budget where forgetting actually happens. We note that this is an argument about *where the budget is spent*, not a claim that weighting wins: the theory does not predict an accuracy advantage for the weighted variant (Sec. 4.6 states the caveat explicitly), and Sec. 5.3 reports what our experiments show.

3.3. FOLoRA

We now describe FOLoRA. For each adapted layer, let $\theta \in \mathbb{R}^d$ denote the flattened adapter parameters (the LoRA matrices A and B, so $d = r(d_{\text{in}} + d_{\text{out}})$), and let $g_\theta = \nabla_\theta \log p(y|x)$ denote the gradient of the log-likelihood with respect to these parameters.

Importance estimation. After finishing task $\mathcal{T}_\tau$, we estimate a per-layer *importance kernel* from the per-sample parameter-gradient autocorrelation,

$$F_\tau \;=\; \mathbb{E}_{(x,y)\sim\mathcal{D}_\tau}\big[\, g_\theta(x,y)\, g_\theta(x,y)^\top \,\big] \;\in\; \mathbb{R}^{d\times d}, \tag{4}$$

which is the empirical Fisher information of task τ restricted to the adapter parameters — i.e. the adapter block $\mathcal{F}_\tau^{\theta\theta}$ of the full-parameter Fisher of Sec. 4 — and which we write F_τ for short below. To keep the estimate tractable we never materialize the full $d\times d$ matrix; we collect the per-sample gradients into a matrix (one row per sample) and retain only its top-k singular directions, i.e., a rank-k approximation of F_τ. We accumulate these kernels across tasks,

$$\bar{F} \;=\; \sum_{\tau<t} F_\tau. \tag{5}$$

Eq. (5) is the *object* the protected subspace is meant to represent. The implementation stores, per layer, a rank-k factor rather than the full F_τ, so what is actually accumulated is computed on the concatenation of the retained factors of all previous tasks; this is an incremental-PCA approximation to the sum in Eq. (5) rather than the sum itself, and it is exact only when the retained top-k subspace of the concatenated factors coincides with the union of the per-task top-k subspaces. We keep the exact form in the equations for readability and use the approximation in all experiments; the two agree in the limit $k \to d$.

Protected subspace. At the start of task $\mathcal{T}_t$, we eigendecompose the accumulated kernel $\bar{F} = \sum_j \sigma_j u_j u_j^\top$, where $\{\sigma_j\}_j$ are the eigenvalues sorted in decreasing order and $\{u_j\}$ the corresponding orthonormal eigenvectors in $\mathbb{R}^d$. We retain the top-k eigendirections as an *importance-weighted protected subspace* $\mathcal{P}_t = \{(\sigma_j, u_j)\}_{j=1}^k$.

Orthogonalization regularizer. Let θ^{t-1} denote the adapter parameter snapshot recorded after task $\mathcal{T}_{t-1}$, and let $\delta\theta = \theta - \theta^{t-1}$ be the parameter displacement accumulated while training task $\mathcal{T}_t$. During the training of task $\mathcal{T}_t$, FOLoRA minimizes

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{CE}}(\theta) \;+\; \lambda \underbrace{\sum_{j=1}^{k} \sigma_j \left(u_j^\top \delta\theta\right)^2}_{\Omega(\delta\theta)}, \tag{6}$$

where $\lambda > 0$ controls the strength and each $u_j \in \mathbb{R}^d$ is a protected direction in the same parameter space as $\delta\theta$. The term $(u_j^\top \delta\theta)^2$ penalizes the alignment of the *displacement* $\delta\theta = \theta - \theta^{t-1}$—rather than the absolute θ—with the protected direction u_j, weighted by the importance σ_j. Using the displacement is essential: at the start of task $\mathcal{T}_t$ we have $\delta\theta = 0$, so the regularizer vanishes and the adapter's existing alignment with past tasks is preserved; only *new* movement into protected directions is penalized. Penalizing the absolute θ would instead push the adapter away from the past tasks' directions and actively erase them, defeating the purpose of protection. Since the regularizer is a differentiable function of A and B, it integrates with standard backpropagation and requires no hard projection, no replay buffer, and no task identity at inference.

Why importance weighting matters. Setting all $\sigma_j = 1$ reduces Ω to an equal-weight orthogonalization, which is the geometry underlying O-LoRA-style orthogonal subspace methods (Sec. 4.6 states the relation exactly). In contrast, FOLoRA's eigenvalue weighting (i) strongly suppresses reuse of high-curvature directions, which is where forgetting is concentrated, and (ii) leaves low-curvature directions nearly free, so the adapter can reuse them for new tasks without wasting rank. This is the mechanism by which FOLoRA spends a fixed rank budget on the directions the bound identifies as costly. We note that the theory does not by itself predict an accuracy advantage for weighting over equal weighting — a larger penalty family is not automatically a better one, and Sec. 4.6 says so explicitly; whether it helps in practice is an empirical question, and Sec. 5.3 reports what our experiments show.

Fig. 1 shows the procedure end to end: the forward pass of one task on the solid row, and the one-time update that produces the penalty for the next task on the dashed row. Two properties of that picture are the ones the rest of the paper relies on: the adapter and its parameter count are shared across all tasks, so nothing in the solid row grows with T; and the quantity that is updated between tasks is the importance state $\{(\sigma_j, u_j)\}$, so the memory FOLoRA spends is an importance-model cost rather than a parameter cost (quantified in Table 4). The same procedure is given in pseudocode as Algorithm 1.

3.4. Relation to prior methods

- **EWC** [3] penalizes per-parameter quadratic displacement $\sum_i F_{ii}(\theta_i - \theta_i^*)^2$ using the *diagonal* Fisher. FOLoRA instead uses the adapter block of the second-order expansion and applies it as a rank-k subspace pro-

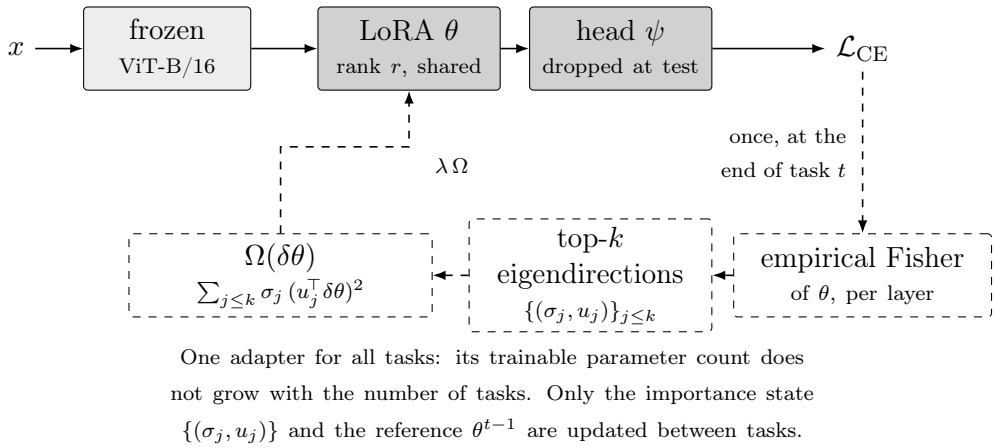


Figure 1: FOLoRA. **Solid row**: the forward pass of task t, where a single shared rank-r adapter θ is trained on a frozen ViT-B/16 backbone with the penalty $\lambda\Omega$ added to the cross-entropy loss; the head ψ is trained but discarded, since prediction is by nearest class mean in the frozen feature space. **Dashed row**: once, after task t ends, the adapter's empirical Fisher is re-estimated per layer, its top k eigendirections are extracted, and they are accumulated into the kernel $\bar{F}$ defining Ω for the next task. The adapter's rank, hence its trainable parameter count, is fixed for the whole sequence; only the importance state grows, so the memory cost is an importance-model cost, not a parameter cost (Table 4).

jection rather than a pointwise penalty. The distinction is not that one is exact and the other is not — both truncate the curvature, EWC to its diagonal and FOLoRA to its top-k spectrum — but that they discard different things: EWC keeps all d coordinates and drops every correlation among them, whereas FOLoRA keeps the k directions of largest curvature with their correlations intact and drops the rest. Which truncation suits a fixed-rank adapter is an empirical question, answered for this setting by the comparison in Sec. 5.2.

- **O-LoRA** [8] allocates a new adapter per task and initializes it orthogo-

Algorithm 1 FOLoRA for class-incremental continual learning

Input: frozen backbone; LoRA rank r; strength λ; budget k.
Initialize shared LoRA adapters $\Delta W^{(l)} = \frac{\alpha}{r} B^{(l)} A^{(l)}$ for each adapted layer l.
$\bar{F}^{(l)} \leftarrow \mathbf{0}$ for all l.
for each task $\mathcal{T}_t$, $t = 1, \ldots, T$ **do**
  if $t > 1$: eigendecompose $\bar{F}^{(l)} = \sum_j \sigma_j u_j u_j^\top$, keep top-$k$ $\{(\sigma_j, u_j)\}$.
  Train adapter parameters $\theta^{(l)}$ and the grown classifier head by minimizing $\mathcal{L}_{\mathrm{CE}} + \lambda \sum_l \sum_j \sigma_j \big(u_j^\top \big(\theta^{(l)} - \theta_{t-1}^{(l)}\big)\big)^2$.
  Record adapter snapshot $\theta_t^{(l)} \leftarrow \theta^{(l)}$.
  Estimate $F_t^{(l)} = \mathbb{E}[g_\theta \, g_\theta^\top]$ on $\mathcal{D}_t$; update $\bar{F}^{(l)} \leftarrow \bar{F}^{(l)} + F_t^{(l)}$.
end for
Output: a single shared adapter with fixed parameter budget.

nally to previous tasks' input subspaces, treating all directions equally; FOLoRA keeps a *single* fixed-budget adapter and protects directions in proportion to their Fisher importance.

- **GPM** [44] and related null-space methods project gradients onto a null space; FOLoRA realizes a *soft*, importance-weighted orthogonal projection that avoids the rank saturation of hard projections.

4. Theoretical Analysis

We derive a second-order bound on the forgetting of a previously learned task under sequential low-rank adaptation, with an explicit remainder term, and show that the FOLoRA regularizer minimizes the leading term of that

bound. We then relate the bound to the equal-weight orthogonalization used by O-LoRA-style methods.

We present the bound as a *design principle rather than a numerical predictor*. We do not claim it is tight, and we never use it to predict how much forgetting a given λ removes. Sec. 4.4 audits the three quantities the expansion rests on — the cubic remainder, the stationarity residual, and the calibration of the Fisher estimate — and reports that the first two are the same order as the quadratic term rather than negligible, and that the third is not a calibrated surrogate for the measured curvature at the operating point we probe. What the theory does establish, and what the method is built on, is structural: the leading term of the expansion is a curvature-weighted projection of the adapter displacement onto the past tasks' eigendirections, and Eq. (15) shows that FOLoRA's penalty is the Lagrangian form of that quantity. The *empirical* case for the method is made separately, by the λ=0 anchor in Sec. 5.3, and not by the bound.

4.1. Setup and assumptions

Let $\theta_\tau \in \mathbb{R}^d$ denote the *adapter* parameters of one adapted layer after learning task $\mathcal{T}_\tau$ — the entries of that layer's low-rank factors A, B, flattened, so that $d = r(d_{\text{in}}+d_{\text{out}})$ — and let ψ_τ denote the classifier-head parameters at that point. Every adapted layer is treated identically, so we carry the analysis for a single layer and drop the layer index; the total adapter term is the sum over layers of the per-layer terms derived below. We write $\xi_\tau = (\theta_\tau, \psi_\tau)$ for the full trainable vector. For a later task $\mathcal{T}_t$ $(t > \tau)$, let

$$\delta\theta^\tau = \theta_t - \theta_\tau, \qquad \delta\psi^\tau = \psi_t - \psi_\tau, \qquad \delta\xi^\tau = (\delta\theta^\tau, \delta\psi^\tau) = \xi_t - \xi_\tau \tag{7}$$

be the adapter, head and full-parameter displacements accumulated *between* τ and t, respectively. We measure the forgetting of task τ by the increase of its empirical loss,

$$\Delta\mathcal{L}_\tau \;=\; \mathcal{L}_\tau(\xi_t) - \mathcal{L}_\tau(\xi_\tau). \tag{8}$$

Our analysis is built on the following assumptions, which we state explicitly because each one is an approximation that the bound inherits.

Assumption 1 (Task stationarity)**.** *ξ_τ is a stationary point of the* unregularized *task loss, $\nabla_\xi \mathcal{L}_\tau(\xi_\tau) = 0$.*

Assumption 2 (Hessian regularity)**.** *$\mathcal{L}_\tau$ is twice continuously differentiable on the segment $[\xi_\tau, \xi_t]$, and its Hessian is Lipschitz there with constant L_3: $\|H_\tau(\xi) - H_\tau(\xi_\tau)\|_2 \le L_3\|\xi - \xi_\tau\|$.*

Assumption 3 (Fisher approximation)**.** *The Hessian at the stationary point is approximated by the empirical Fisher, $H_\tau(\xi_\tau) \approx \mathcal{F}_\tau$,* assumed *rather than derived, following the standard practice of the EWC family [3, 4]. We state it as an assumption because the Gauss–Newton equivalence holds for the* true *Fisher at a maximum-likelihood point under a correctly specified model [70], whereas the empirical Fisher estimated here from a finite per-sample budget is in general a different matrix and can deviate substantially from the Hessian outside that regime [71]. The consequences of this substitution are not absorbed silently: they appear explicitly as ε_{F} in Eq.* (9)*, which is by construction the exact error of the substitution at the given displacement.*

Defining ε_{F} as the *exact* error rather than an upper bound on it has a consequence worth stating: Eq. (9) does not actually require Assumption 3 in order to be valid — since $\Delta\mathcal{L}_\tau = \frac{1}{2}\delta\xi^{\tau\top}\mathcal{F}_\tau\delta\xi^\tau + \frac{1}{2}\delta\xi^{\tau\top}(H_\tau(\xi_\tau) - \mathcal{F}_\tau)\delta\xi^\tau +$

remainder, the inequality holds for the empirical Fisher as computed, independently of its quality. The assumption governs the *usefulness* of the bound rather than its validity: it is what licenses treating ε_{F} as small and $\mathcal{F}_\tau$ as an object worth regularizing against.

Assumption 1 is the same assumption used by EWC [3] and GPM [44]. It applies to the *unregularized* task loss: FOLoRA's training objective is $\mathcal{L}_{\mathrm{CE}} + \lambda\Omega$, so at the end of task $\mathcal{T}_\tau$ we have $\nabla_\xi \mathcal{L}_{\mathrm{CE}}(\xi_\tau) = -\lambda \nabla_\xi \Omega(\theta_\tau)$ rather than exactly zero. We discuss the size of this deviation in Remark 2 and quantify it empirically in Sec. 5.4.

4.2. A second-order forgetting bound

Theorem 1 (Second-order forgetting bound)**.** *Under Assumptions 1–2, the forgetting of task $\mathcal{T}_\tau$ after learning task $\mathcal{T}_t$ satisfies*

$$\Delta\mathcal{L}_\tau \;\le\; \frac{1}{2}\,\delta\xi^{\tau\top}\mathcal{F}_\tau\,\delta\xi^{\tau} \;+\; \frac{L_3}{6}\big\|\delta\xi^{\tau}\big\|^3 \;+\; \varepsilon_{\mathrm{F}}, \tag{9}$$

where $\varepsilon_{\mathrm{F}} = \frac{1}{2}\big|\delta\xi^{\tau\top}(H_\tau(\xi_\tau) - \mathcal{F}_\tau)\delta\xi^{\tau}\big|$ is the absolute error incurred by replacing the Hessian with the Fisher (Assumption 3), and $\mathcal{F}_\tau = \mathbb{E}_{(x,y)\sim\mathcal{D}_\tau}[\, g_\xi g_\xi^\top]$, with $g_\xi \equiv \nabla_\xi \log p_\xi(y|x)$, is the empirical Fisher of task τ over all trainable parameters.

Proof. Taylor-expanding $\mathcal{L}_\tau$ at ξ_τ along the displacement $\delta\xi = \xi_t - \xi_\tau$ with the integral form of the remainder gives

$$\mathcal{L}_\tau(\xi_t) = \mathcal{L}_\tau(\xi_\tau) + \nabla\mathcal{L}_\tau(\xi_\tau)^\top\delta\xi + \int_0^1 (1-s)\,\delta\xi^\top H_\tau(\xi_\tau + s\,\delta\xi)\,\delta\xi\,\mathrm{d}s. \tag{10}$$

By Assumption 1 the linear term vanishes. Decomposing the integral,

$$\Delta\mathcal{L}_\tau = \frac{1}{2}\,\delta\xi^\top H_\tau(\xi_\tau)\,\delta\xi + \int_0^1 (1-s)\,\delta\xi^\top\big[H_\tau(\xi_\tau{+}s\delta\xi) - H_\tau(\xi_\tau)\big]\delta\xi\,\mathrm{d}s. \tag{11}$$

The integrand is bounded by $(1-s)\,\|\delta\xi\|^2\,\|H_\tau(\xi_\tau+s\delta\xi)-H_\tau(\xi_\tau)\|_2 \le (1-s)\,s\,L_3\|\delta\xi\|^3$ by Assumption 2, and $\int_0^1(1-s)s\,\mathrm{d}s = 1/6$, which bounds the remainder by $\frac{L_3}{6}\|\delta\xi\|^3$. Substituting the Fisher approximation (Assumption 3) into the leading term yields Eq. (9). □

Isolating the adapter block. Because $\xi = (\theta, \psi)$ is a concatenation, the leading term of Eq. (9) splits into an adapter block, a head block and a cross term. Writing $\mathcal{F}_\tau$ in the same block structure,

$$\tfrac{1}{2}\,\delta\xi^{\tau\top}\mathcal{F}_\tau\,\delta\xi^\tau = \underbrace{\tfrac{1}{2}\,\delta\theta^{\tau\top}F_\tau\,\delta\theta^\tau}_{\text{adapter block}} + \underbrace{\delta\theta^{\tau\top}\mathcal{F}_\tau^{\theta\psi}\delta\psi^\tau + \tfrac{1}{2}\,\delta\psi^{\tau\top}\mathcal{F}_\tau^{\psi\psi}\delta\psi^\tau}_{\text{head / cross terms}}, \quad F_\tau := \mathcal{F}_\tau^{\theta\theta}, \tag{12}$$

so that Eq. (9) becomes

$$\Delta\mathcal{L}_\tau \;\le\; \underbrace{\frac{1}{2}\,\delta\theta^{\tau\top}F_\tau\,\delta\theta^\tau}_{\text{adapter term}} + \underbrace{\frac{L_3}{6}\big\|\delta\xi^\tau\big\|^3}_{\text{higher order}} + \varepsilon_{\mathrm{F}} \;+\; R_{\text{head}}, \tag{13}$$

where R_{head} collects the two head/cross terms of Eq. (12) and F_τ is the adapter block of $\mathcal{F}_\tau$ — the empirical Fisher restricted to the adapter parameters of one layer, of size $d \times d$. FOLoRA constrains the first term of Eq. (13) and nothing else; we carry R_{head} explicitly rather than dropping it, because the classifier head is trained between τ and t and its contribution to $\Delta\mathcal{L}_\tau$ is not zero.

Three approximations are made in *estimating* F_τ, as opposed to in the bound itself, and we state all three. First, the estimate is a rank-k truncation, as described in Sec. 3. Second, it is computed per adapted layer and treated as block-diagonal across layers: correlations between the parameters of different adapted layers are not modelled, so the $d \times d$ block above is one

diagonal block of the full adapter Fisher of dimension $L\,r(d_{\text{in}} + d_{\text{out}})$ for L adapted layers. Third, the weights are on the scale of an *unnormalized* estimate. The per-direction weight σ_j^2 carried by the released implementation is an eigenvalue of $G^\top G$ for the matrix G of per-sample adapter gradients accumulated over the Fisher budget, and therefore grows with the number of samples in that budget (300 at our default), whereas the diagonal Fisher of the EWC baseline we compare against is explicitly averaged over its samples. A numerical value of λ is consequently meaningful only relative to the convention its own method uses: the λ grids of Sec. 5 are read in that sense, and we do not compare λ across the two families on an absolute scale. All three are properties of the estimator; the bound is stated for the exact F_τ, so none weakens Theorem 1, but all three are part of what a reader comparing the paper to the released code should expect to find. The decomposition is useful only if the adapter term dominates, which holds when the displacement is small. In our setting this is not merely a hope: the adapter has a fixed rank $r = 16$ per layer and is trained under an explicit penalty on its displacement, so $\|\delta\theta^\tau\|$ is bounded by the geometry of the problem rather than growing with the number of tasks. Whether the higher-order term is in fact negligible at that displacement is a question about the data, not about the algebra, and we measure it rather than assume it: Sec. 4.4 finds it to be the *same order* as the quadratic term, which is why Eq. (13) is described throughout as an upper bound with quantified slack rather than as a tight characterization of forgetting.

Eq. (13) is a bound on the *loss* of task τ, and in that bound the classifier head contributes through R_{head}. Our evaluation, however, does not use the

head: under the NCM protocol (Sec. 5.1) the head is discarded at test time and prediction is made by nearest class mean in the frozen feature space, so the adapter is the *only* pathway by which a task's test accuracy can be affected after it has been learned. The bound's scope and the evaluation protocol therefore coincide in the sense that matters for the reported metric, even though they differ in the loss that the bound is stated for.

Remark 1 (Scope: one reference point)**.** *Summing Eq.* (13) *over all previous tasks* $\tau < t$ *requires a displacement* $\delta\theta^\tau$ per *task* τ*, i.e. a snapshot of the adapter parameters at every past task. FOLoRA stores a single reference* θ^{t-1}*, following the same single-reference approximation as EWC [3], and replaces* $\delta\theta^\tau$ *by* $\delta\theta = \theta_t - \theta^{t-1}$*. The induced error is* $\theta^{t-1} - \theta^\tau$*, i.e. exactly the displacement accumulated between tasks* τ *and* $t-1$ *— which is itself penalized by the same regularizer at every intermediate task. Writing* R *for the head terms* R_{head} *summed over* τ*, the bound we actually minimize is*

$$\begin{aligned}\sum_{\tau<t} \Delta\mathcal{L}_\tau \;\le\; \frac{1}{2}\,\delta\theta^\top\Big(\sum_{\tau<t} F_\tau\Big)\delta\theta \;+\; (t-1)\Big(\frac{L_3}{6}\|\delta\xi^{\max}\|^3 + \varepsilon_{\mathrm{F}}\Big) \\ +\; R \;+\; \mathcal{O}\big(\|\theta^{t-1} - \theta^\tau\|^2\big),\end{aligned} \tag{14}$$

where we identify $\sum_{\tau<t} F_\tau$ *with the accumulated importance kernel* $\bar{F}$ *of Eq.* (5)*. The scope of the theorem is thus "forgetting relative to the most recent snapshot", and the statement degrades gracefully rather than being exact for all historical tasks.*

Remark 2 (Deviation from stationarity under the regularizer)**.** *Because FOLoRA minimizes* $\mathcal{L}_{CE} + \lambda\Omega$*, the stationarity Assumption 1 holds only approximately, with residual gradient* $-\lambda\nabla_\xi\Omega(\theta_\tau)$ *— nonzero only*

in the adapter block, since Ω does not depend on the head. Writing $\Omega(\delta\theta) = \sum_j \sigma_j (u_j^\top \delta\theta)^2$ gives $\nabla_\theta \Omega(\theta_\tau) = 2\bar{G}^\top \bar{G}\, \delta\theta$ where $\bar{G}$ stacks the accumulated Fisher directions. The residual therefore vanishes proportionally to the displacement itself: in directions where protection is effective ($\delta\theta \to 0$) the assumption is self-consistently satisfied, while in weakly protected directions it degrades linearly in λ. This provides a theoretical counterweight to increasing λ beyond the range where the displacement is small, and is consistent with the ablation in Sec. 5.3: at k=16, where the penalty acts on fewer directions, average accuracy peaks at $\lambda \in \{3, 10\}$ and then declines through $\lambda = 100$ before saturating by $\lambda = 300$. At k=64 the same decline is not visible over the range we probed, and we do not claim it there.

4.3. FOLoRA minimizes the bound

Using the eigendecomposition $\bar{F} = \sum_j \sigma_j u_j u_j^\top$, the leading term of Eq. (14) becomes

$$\frac{1}{2}\, \delta\theta^\top \bar{F}\, \delta\theta \;=\; \frac{1}{2} \sum_j \sigma_j \left(u_j^\top \delta\theta\right)^2, \tag{15}$$

where $\delta\theta \in \mathbb{R}^d$ is the adapter-parameter displacement and each $u_j \in \mathbb{R}^d$ is a unit eigenvector of $\bar{F}$, so every inner product above is between vectors of the same dimension. Equation (15) shows that the dominant term of the forgetting bound is a sum of squared projections of the displacement onto the eigendirections of the accumulated Fisher, weighted by their importance σ_j. Here σ_j denotes an eigenvalue of $\bar{F}$; the implementation stores each direction scaled by the corresponding singular value $s_j = \sqrt{\sigma_j}$ of the per-sample gradient matrix, so that the squared norm it penalizes carries weight

σ_j exactly as written. The FOLoRA regularizer in Eq. (6),

$$\Omega(\delta\theta) = \sum_j \sigma_j \left(u_j^\top \delta\theta\right)^2, \qquad \delta\theta = \theta - \theta^{t-1}, \tag{16}$$

is exactly the penalty term of the Lagrangian relaxation of the constraint that the leading term of the bound remain below a tolerance $\epsilon > 0$,

$$\delta\theta^\top \bar{F}\, \delta\theta \;\leq\; \epsilon, \tag{17}$$

since relaxing that constraint with multiplier λ returns $\mathcal{L}_{\mathrm{CE}} + \lambda\Omega$. Minimizing Ω therefore drives $|u_j^\top \delta\theta|$ toward zero for the largest σ_j, which in turn minimizes the dominant terms of Eq. (15). We therefore interpret FOLoRA as the algorithm that *directly minimizes the leading term of a second-order bound on catastrophic forgetting*, with λ controlling the stability–plasticity trade-off and k controlling the rank of the protected subspace.

4.4. How tight is the bound?

Theorem 1 is an upper bound, and everything above uses only its *structure*. We do not assume it is tight, and we never use it to predict how much forgetting a given λ removes. To make that a measured statement rather than a disclaimer, we probed the bound's own terms on a trained run; the probe protocol, the decision rule frozen before the numbers were computed, and the full measurements are in Appendix Appendix B. Three of the approximations the expansion inherits are violated at the operating point we probe:

(i) The remainder is not negligible. The cubic coefficient is of the same order as, and at its worst boundary 2.5×, the quadratic term it is meant to correct. *(ii) The stationarity residual is not small either.* It does not decay

fast enough to be dropped, so the "quadratic plus corrections" reading of the expansion is not numerically available at this λ. *(iii) The finite-sample Fisher estimate is not a calibrated surrogate for the curvature.* It underestimates the measured curvature along the realized displacement by four to five orders of magnitude — which does *not* contradict Theorem 1, stated as it is for the exact F_τ, but does mean that Assumption 3 is far from innocuous here, which is precisely the regime the empirical-Fisher literature warns about [71]. The operative reading is the same in all three cases: the bound's leading term is not a quantitative predictor of forgetting in this setting.

What this licenses. The bound is used here as a *design principle*. It identifies which quantity a regularizer should suppress — the curvature-weighted projection of the adapter displacement onto the past tasks' protected directions — and Eq. (15) shows that FOLoRA's penalty is exactly the Lagrangian form of that quantity. We make no claim that minimizing it produces a predictable reduction in forgetting, and Sec. 5 does not test such a claim. The empirical case for FOLoRA is made by ablation instead, and it is a clean one: removing Ω from the objective and changing nothing else takes Split CIFAR-100 accuracy from 78.32 to 69.59 (n=5, paired $p = 0.0005$, negative in all five seeds), i.e. down to the level of unprotected sequential fine-tuning (69.31, $p = 0.53$). The entire effect of the method is carried by the term the bound identifies, which is the sense in which the derivation is load-bearing — as motivation for *what* to penalize, not as a guarantee of *how much* it helps.

4.5. Relation to prior forgetting bounds

Second-order analyses of catastrophic forgetting are well established for models trained from scratch. Nguyen et al. [55] show that forgetting is driven by the alignment between successive tasks' loss landscapes and derive a worst-case bound in terms of the curvature of the old task; Evron et al. [56] give tight bounds for linear networks showing that forgetting is governed by the overlap between the old task's input distribution and the directions in which the new task changes the weights; Karakida and Akaho [57] analyse learning curves for continual learning using NTK overlap matrices; and Doan et al. [58] quantify forgetting through the NTK overlap between tasks. Closer in spirit, orthogonal-gradient descent [46, 43] admits generalization bounds obtained by constraining updates to the null space of previous task gradients.

Our analysis differs from this line in where it is instantiated. Prior results bound forgetting for *full* parameter updates of models trained from scratch, and are used mainly to explain forgetting. We instantiate the bound in the *low-rank adapter subspace* of a frozen pre-trained backbone, where the displacement is supported on a $r(d_{\text{in}} + d_{\text{out}})$-dimensional subspace per layer, and we use the bound constructively: its dominant term is exactly the quantity a subspace-regularizer can minimize, which yields the FOLoRA objective. The resulting penalty differs from null-space projection [44, 46] in being importance-weighted and soft rather than hard, so that the fixed rank budget is spent on the directions with the largest curvature.

4.6. Connection to equal-weight orthogonalization

If all eigenvalues are set to $\sigma_j = 1$, the leading term of the bound, Eq. (15), becomes $\frac{1}{2}\|\bar{U}^\top \delta\theta\|^2$, where $\bar{U} = [u_1, \ldots, u_k]$ is the orthonormal basis of the re-

tained protected subspace — the equal-weight geometry underlying O-LoRA-style orthogonal subspace methods. The eigenvalue-weighted family therefore contains the equal-weight geometry as the special case $\sigma_j \equiv 1$, and the two coincide exactly when the spectrum is flat over the protected subspace; on our checkpoints it is not flat, so the weighted family is the strictly larger one as a family of penalties.

A larger family is not by itself an accuracy claim, and one tempting over-claim is worth pre-empting here: because every coefficient in Eq. (15) is non-negative, weighting by σ_j does *not* reduce the value of the bound at a fixed displacement – the weighted coefficients can exceed 1. The claim is the weaker and well-defined one that the weighted penalty attains a *lower value of the bound after both objectives are optimized* for a given protected rank k, by spending the fixed rank budget where the curvature is largest. Whether that is realized in practice is an empirical question; Sec. 5.3 characterizes the weighted family over λ and k, and the equal-weight special case $\sigma_j \equiv 1$ is its natural control.

5. Experiments

5.1. Setup

We use a ViT-B/16 backbone pre-trained on ImageNet-1k, frozen throughout, with LoRA adapters (rank $r = 16$, $\alpha = 16$) attached to the MLP down-projection of every transformer block (12 adapters in total). The classifier head is a single linear layer grown as new classes arrive. All methods share the identical backbone and head; they differ only in the adapter regularization.

Datasets. Split CIFAR-100 (20 tasks $\times$ 5 classes) and Split ImageNet-R (20 tasks $\times$ 10 classes), both under the class-incremental protocol (task identity unavailable at test time).

Baselines. (i) *Seq-LoRA*: sequential fine-tuning of the shared adapter with no protection (lower bound); (ii) *EWC-LoRA*: diagonal Fisher on the LoRA parameters [3]; (iii) *O-LoRA*: per-task adapters with orthogonal initialization [8]; selected per input by key–query matching [6]; (v) *CODA-Prompt*: a set of prompt components combined by attention [7]; (vi) *InfLoRA*: a per-task subspace projected to interfere minimally with the previous ones [19]; (vii) *FOLoRA* (ours).

Training. AdamW with learning rate 10^{-3}, batch size 32, and bf16 mixed precision. We train the adapter-based methods (Seq-LoRA, EWC-LoRA, O-LoRA, InfLoRA, FOLoRA) for 5 epochs per task — for the Fisher-weighted ones this avoids over-fitting and keeps the empirical Fisher meaningful (Sec. 5.4), and it places the whole adapter family on one budget — the prompt-based methods (L2P, CODA-Prompt) are trained for 20 epochs per task, their optimal setting since their key-based prompt selection benefits from more epochs. Prompt geometry is a substantive choice rather than a detail — CODA-Prompt's contribution is a *large* pool of components combined by attention — so we use each prompt method's own geometry (L2P: pool 20, length 5; CODA-Prompt: pool 100, length 8) and report CODA-Prompt at its faithful geometry. We also ran CODA-Prompt with L2P's geometry (pool 20, length 5) at 20 epochs; it scores 75.83 on CIFAR-100 against the faithful 74.71. We report the faithful configuration, i.e. the *lower* of the two numbers we measured, so the CODA-Prompt margin quoted be-

low is the smaller of the two available to us. For FOLoRA we fix the main configuration by a grid search and then do not retune it: $\lambda \in \{1, 3, 10\}$ at k=64 and $k \in \{16, 64, 128\}$ at λ=3, plus λ=0 (penalty removed, everything else unchanged) as the ablation control. The main configuration k=64, λ=3 was fixed *before* the extra seeds of the top-up landed and is reported as such throughout; λ=10 is not distinguishable from it on the five seeds the two share ($\Delta = +0.08$, $p = 0.887$, Table 3) and appears as a grid point there, not as the main configuration. The importance kernel is estimated from up to 300 per-sample gradients per task. EWC's regularization strength is likewise tuned on CIFAR-100: $\lambda \in \{1, 3, 10, 30\}$ at five seeds, and a wider grid $\lambda \in \{100, 300, 1000, 3000\}$ at ten seeds that was run on both benchmarks (λ=3000 was not run on ImageNet-R). In Table 1 we report each benchmark's *best-scoring* EWC point from that grid — λ=300 on CIFAR-100 and λ=1000 on ImageNet-R — i.e. we deliberately take the strongest configuration of the baseline *on each benchmark*, and we report the whole curve, including the pre-registered baseline λ=100, in Table 2.

Metrics and evaluation protocol. We follow the frozen-feature nearest-class-mean (NCM) protocol standard in this literature [6, 7, 19]: after the final task we L2-normalize the CLS feature of the adapted backbone and classify by the nearest class mean computed from the training features of each task seen so far, with each task's means computed from that task's own training split. This is the protocol under which L2P, CODA-Prompt and InfLoRA are reported, which makes the comparison directly interpretable; it also removes the classifier head from the measurement, so the reported numbers isolate the quality of the adapted representation rather than the

growth of the head. We report average accuracy over all tasks after the final task ($\overline{\text{ACC}}$) and the CIL–TIL gap (GAP). GAP is the mean drop, in points, when a task's test set is classified against all seen classes rather than against that task's own classes, under the same final-model feature extraction: it measures how far a method's decision rule is from relying on task identity, and lower is better. Because every method is run on the same seed set—and therefore sees the same class order and initialization—we assess the significance of differences between methods with a paired t-test over the seeds common to both. Seed budgets are *not* equal across methods and we do not pool them: EWC-LoRA and FOLoRA have $n = 10$ on both benchmarks; Seq-LoRA and O-LoRA $n = 10$ on CIFAR-100 and $n = 5$ on ImageNet-R; InfLoRA 5/3; L2P 5/5; CODA-Prompt 5/3. SimpleCIL is a deterministic floor ($n = 1$, no variance) and is not tested. Every p-value below states the common-seed count it was computed on, and Table 1 repeats the budget for each method.

Two exemplar-free methods are adjacent enough to warrant a word. FeCAM [72] and EFC [73] are among the closest exemplar-free continual learners to the setting studied here, and neither is a baseline of Table 1 — for reasons that are properties of our protocol rather than judgements of their quality. FeCAM contributes a *classifier*: it fits anisotropic class prototypes to frozen features, which is exactly the regime our NCM head occupies, so adding it would replace the evaluation rule that every row currently shares and would therefore measure a classifier rather than the training-time regularizer this paper varies. EFC contributes a mechanism that requires the backbone *not* to be frozen — its feature-drift regularizer exists to hold a mov-

ing backbone's features stable — so running it under frozen features would delete the mechanism under test. Holding the classifier and the backbone fixed across every method is what makes the comparisons in Table 1 isolate the training procedure; we accordingly claim nothing against FeCAM or EFC as complete systems, and comparing against either would call for a different experimental design than the one used here. Significance is reported on Holm–Bonferroni corrected p-values, with the uncorrected value given alongside whenever a claim turns on the difference. Table 1 fixes the confirmatory family, because it asks one question per baseline: within a benchmark and a metric, the contrasts against the nine CIFAR-100 and eight ImageNet-R baselines are one family, adjusted step-down over the sorted p-values. We do not pool the two benchmarks into a single family of seventeen tests. They answer different questions—on CIFAR-100 the comparison is to a baseline at parity, on ImageNet-R it is not—and pooling would inflate the family to buy protection against an error mode that neither table introduces. Elsewhere—the EWC λ grid of Table 2, the ablations of Sec. 5.3, the O-LoRA variants of Sec. 5.4—the comparison is exploratory and the family is stated where the number is used. The correction removes no claim the argument rests on. The verdict pre-registered before any seed was run, ImageNet-R against EWC-LoRA at $\lambda = 100$, survives at $p_{\text{Holm}} = 0.0028$ on forgetting and 0.0011 on average accuracy; its CIFAR-100 counterpart was never claimed as significant and remains so ($p = 0.0734$, corrected 0.2934); and the paper's largest effect, the λ=0 ablation, survives a family more than three times larger than the confirmatory one ($p = 0.0005$, corrected 0.011 across all 28 CIFAR-100 configurations). What the correction does remove are marginal forgetting

differences against L2P, CODA-Prompt and the weaker EWC-LoRA operating points ($0.023 \leq p \leq 0.040$); none is claimed below, and each is named where it arises. The correction now admits one forgetting contrast it previously did not, against O-LoRA, which runs in FOLoRA's favour once every per-task adapter is evaluated.

5.2. Main results

The protocol-independent facts hold across both tables: all methods share one frozen backbone and one classifier head. What separates the families is the per-task budget: FOLoRA, Seq-LoRA, EWC-LoRA, L2P and CODA-Prompt reuse one fixed set of trainable parameters across tasks, whereas O-LoRA and InfLoRA allocate a new adapter per task and therefore grow linearly with the number of tasks (Table 4). The classifier head grows with the number of classes seen for *every* method, because the class-incremental protocol requires it and all methods share it; it is trained but discarded at evaluation under the NCM protocol. That is why the trainable-parameter totals in Table 4 rise for every row between T=1 and T=20 while the adapter budget of the five methods above does not, and the table separates the two. Fig. 2 and Fig. 3 take the same final representations and re-score them as the candidate class set grows; we introduce them together with the semantics of that axis, because it is easy to misread as a training trajectory. Each figure shows the candidate-set restriction curves that the corresponding column of Table 1 collapses into a single final-task average. Both figures re-score a *single* feature extraction of the terminal model: the horizontal axis is the number of candidate classes the nearest-class-mean decision must choose among (5 per task on CIFAR-100, 10 on ImageNet-R, up to 100 and 200 respectively), *not*

Table 1: Class-incremental results under the frozen-feature nearest-class-mean protocol (mean ± std over seeds; $\overline{\text{ACC}}$ higher is better). GAP is the CIL–TIL gap: the mean drop, in points, when a task's test set is classified against all seen classes rather than its own, under a single final-model feature extraction (lower is better: less reliance on task identity). EWC-LoRA is shown at the best of its tuned points on each benchmark — λ=300 (CIFAR-100); λ=1000 (ImageNet-R) — i.e. the strongest configuration of the baseline, with its full λ curve, including the pre-registered judging baseline λ=100, in Table 2. The accuracy comparison is parity at every point tested *except* ImageNet-R at λ=100 (+2.85, $p_{\text{Holm}} = 0.0011$), where the paired difference reaches significance after correction. On CIFAR-100 the difference changes sign across the baseline's own tuned points, so the ordering of the two methods is not stable there (full curve in Table 2). On the CIL–TIL gap FOLoRA has the *smaller* gap at the 1 tuned point where the difference survives correction, and EWC-LoRA at none; the full curve is in Table 2. Seed budgets are *not* equal: EWC-LoRA and FOLoRA n=10 on both benchmarks; Seq-LoRA and O-LoRA 10/5 (CIFAR-100/ImageNet-R); InfLoRA 5/3; L2P 5/5; CODA-Prompt 5/3; SimpleCIL is the deterministic floor (n=1, no variance, untested). Bold marks the best value *among the rows shown*; ties are not bolded.

	Split CIFAR-100		Split ImageNet-R	
Method	$\overline{\text{ACC}}$	GAP	$\overline{\text{ACC}}$	GAP
Seq-LoRA	69.31±1.92	9.92±0.58	56.58±1.49	8.83±0.70
EWC-LoRA	**78.00±0.54**	**7.35±0.37**	66.33±0.61	**7.00±0.30**
O-LoRA	64.94±3.03	11.03±0.82	53.72±0.65	9.72±0.35
L2P	73.65±0.81	8.12±0.37	59.47±0.43	8.02±0.36
CODA-Prompt	74.71±0.71	7.71±0.55	59.31±0.22	7.83±0.42
InfLoRA	72.95±0.64	9.60±0.44	58.93±0.19	8.86±0.52
SimpleCIL (floor)	70.31±0.00	8.48±0.00	49.71±0.00	7.96±0.00
FOLoRA (ours)	77.87±0.90	7.42±0.59	**66.37±0.77**	7.27±0.49

Table 2: EWC-LoRA regularization sensitivity. Each λ is compared against FOLoRA with a paired t-test over the seeds common to both; n is that common count, *not* either side's own budget, and it is why the FOLoRA column repeats the same value on every row of a benchmark. λ=100 is the baseline fixed in advance, before the seed top-up that closed the grid to n=10; λ=3000 was not run on ImageNet-R. The printed p is uncorrected; * marks $p < 0.05$ *after* Holm–Bonferroni correction within the confirmatory family of Table 1 (Sec. 5.1), so a row may print $p < 0.05$ without a star. The accuracy comparison is parity at every point tested *except* ImageNet-R at λ=100 (+2.85, $p_{\text{Holm}} = 0.0011$), where the paired difference reaches significance after correction. On CIFAR-100 the difference changes sign across the grid (+0.65 at λ=100 to −0.14 at λ=300), so the ordering of the two methods is not stable across the baseline's own tuned points. On the CIL–TIL gap FOLoRA has the *smaller* gap at the 1 tuned point where the difference survives correction, and EWC-LoRA at none: ImageNet-R λ=100 (FOLoRA 7.27 against EWC-LoRA 8.44, $p_{\text{Holm}} = 0.0028$).

			$\overline{\text{ACC}}$			
Benchmark	EWC λ	n	EWC	FOLoRA	Δ	p
CIFAR-100	100	10	77.22±1.08	77.87±0.90	+0.65	0.0734
CIFAR-100	300	10	78.00±0.54	77.87±0.90	-0.14	0.6440
CIFAR-100	1000	10	77.65±0.42	77.87±0.90	+0.21	0.4961
CIFAR-100	3000	10	77.40±0.62	77.87±0.90	+0.47	0.2507
ImageNet-R	100	10	63.52±1.28	66.37±0.77	+2.85	0.0002*
ImageNet-R	300	10	66.32±0.62	66.37±0.77	+0.05	0.9010
ImageNet-R	1000	10	66.33±0.61	66.37±0.77	+0.04	0.9235
ImageNet-R	3000	—	—	66.37±0.77	—	—

elapsed training and *not* the state of the model at that point of the sequence. The leftmost point therefore already contains the forgetting accumulated over the whole sequence, and the curve shows what happens to one fixed representation as the candidate set grows. On CIFAR-100 all six curves lie within five points of each other at the smallest candidate set (93.6 to 98.2) and span 64.9 to 78.0 at the full one; FOLoRA and EWC-LoRA in particular are indistinguishable at every candidate set, their paired difference ranging over only 0.66 points and never exceeding 0.46 in either direction. On ImageNet-R the spread at the smallest candidate set is already 7.8 points (82.9 for CODA-Prompt to 90.7 for EWC-LoRA, with FOLoRA at 90.5 and O-LoRA at 83.2), and FOLoRA's difference from EWC-LoRA stays small as the candidate set grows rather than widening: -0.19 at the smallest set, -1.12 at the fifth task's set, $+0.04$ at the full set, a spread of 1.15 points — which is why we do not read that accuracy difference as an advantage that compounds with the number of tasks.

FOLoRA matches the strongest regularized baseline; it does not beat it. Under the comparison fixed in advance — a paired t-test against EWC-LoRA at $\lambda = 100$ — FOLoRA is ahead by $+0.65$ points on CIFAR-100 ($p = 0.0734$, which the pre-registered rule calls "matches or slightly exceeds" and *not* "outperforms") and by $+2.85$ points on ImageNet-R ($p = 0.0002$, all ten per-seed differences positive), which the same rule calls a significant lead. We report that verdict, and also why it must not be read as a general advantage: λ is a hyperparameter of the *baseline*, and the comparison moves with it. On CIFAR-100 EWC-LoRA's four tuned points span 77.22 to 78.00 while FOLoRA scores 77.87 — inside that span — with paired differences of

-0.14 ($p = 0.644$) at $\lambda = 300$, $+0.21$ ($p = 0.496$) at $\lambda = 1000$ and $+0.47$ ($p = 0.251$) at $\lambda = 3000$. That is parity at every CIFAR-100 point, and EWC-LoRA's best point sits 0.13 points *above* FOLoRA. On ImageNet-R the same structure appears with a wider spread, and the lead is confined to one end of it: $\lambda = 100$ is that benchmark's weakest EWC point at 63.52, and it is there — and only there — that the comparison reaches significance, while at $\lambda = 300$ and $\lambda = 1000$, both roughly 2.8 points higher, the difference collapses to $+0.05$ ($p = 0.901$) and $+0.04$ ($p = 0.924$). The sign of our difference against EWC-LoRA is therefore set by which operating point of the baseline is read, so we report the whole curve (Table 2) rather than the point that reads best for us. EWC-LoRA was not run at $\lambda = 3000$ on ImageNet-R.

On the CIL–TIL gap the comparison is mixed, and we report both directions. The direction is set by the baseline's regularization strength, not by the method. On CIFAR-100 EWC-LoRA's gap is the smaller one at the two larger strengths, but neither difference survives the correction of Sec. 5.1: the paired gap is $+0.32$ points at $\lambda = 1000$ and $+0.39$ at $\lambda = 3000$, nominally significant at $p = 0.0400$ and 0.0401 and corrected to 0.16 in both cases, so on this benchmark the two methods are indistinguishable in the gap direction. At $\lambda = 300$ the difference is $+0.07$ ($p = 0.568$) and at the pre-registered $\lambda = 100$ it is -0.30 ($p = 0.071$), where the sign runs the other way without reaching significance either way. On ImageNet-R the direction reverses and there the difference survives: FOLoRA has the significantly *smaller* gap at $\lambda = 100$ (-1.17 points, $p = 0.0004$, corrected 0.0028). At $\lambda = 300$ the nominal advantage (-0.41, $p = 0.0317$) does not

survive correction (0.13), and at $\lambda = 1000$ the two are indistinguishable ($+0.27$, $p = 0.203$). Because average accuracy and the gap are measured under the same final feature space—a representation that separates the classes better both classifies better and loses less when the candidate set grows—the two co-vary across these methods. We therefore do not treat the gap as an independent axis of merit in either direction; we report it rather than leaving a reader to assume the favourable direction.

FOLoRA is ahead of the parameter-efficient family, and there the difference is uniform. Against L2P, CODA-Prompt, InfLoRA and O-LoRA the paired differences in average accuracy are significant on both benchmarks with every per-seed difference of the same sign: across the four baselines the mean paired difference runs from +3.61 to +12.92 points on CIFAR-100 (corrected $p \leq 0.008$) and from +7.45 to +13.20 on ImageNet-R (corrected $p \leq 0.007$). The upper end of each range is set by the initialization-only O-LoRA row, and it narrows against the training-time orthogonalized objective that the original paper specifies, which we also ran: on CIFAR-100 to +7.53 and +5.83 at the two constraint strengths (λ_1=0.1 and λ_1=1.0; corrected $p = 0.0035$ and 0.0129 within the two-member family those variants form), and on ImageNet-R to +10.59 and +5.74 (corrected $p = 0.034$ each in the same family). This is the paper's positive empirical result. It is also the comparison in which the baselines' seed budgets are smallest ($n = 5$, or $n = 3$ for CODA-Prompt and InfLoRA on ImageNet-R), which is why Sec. 5.1 lists each budget separately instead of implying a shared one — and why the claim is stated against the parameter-efficient family rather than against full-parameter regularization. Two rows fall below the frozen-

feature floor, and it matters which two: O-LoRA at its initialization-only setting on CIFAR-100 (64.94 against a floor of 70.31), and — more tellingly — Seq-LoRA (69.31), which trains the shared adapter sequentially with no protection and ends *below* the frozen-feature floor, i.e. below simply classifying the frozen backbone's own features. The latter is direct evidence that the forgetting this paper targets lives in the adapter itself rather than in the head. Every other row clears the floor: InfLoRA at 72.95 on CIFAR-100 and 58.93 on ImageNet-R, O-LoRA at 53.72 on ImageNet-R, against floors of 70.31 and 49.71 respectively; and the training-time orthogonality term closes the O-LoRA shortfall on CIFAR-100 (70.84 and 72.54); Sec. 5.4 reports those configurations in full.

5.3. Ablation study

We ablate FOLoRA's two hyperparameters — the regularization strength λ and the number of protected directions k — on Split CIFAR-100. Table 3 reports average accuracy and the CIL–TIL gap for each setting together with a paired t-test against the pre-registered main configuration (k=64, λ=3).

The penalty carries the entire effect. Setting $\lambda = 0$ removes the Fisher-weighted orthogonality penalty and changes nothing else; on the five seeds common to both configurations average accuracy falls from 78.32 to 69.59 ($\Delta = +8.72$, $p = 0.0005$), i.e. to the level of unprotected sequential fine-tuning (69.31, Table 1). (The λ=3 full ten-seed mean is the 77.87 of Table 1; the two figures differ only in seed budget, not in configuration.) The benefit of FOLoRA is therefore attributable to the regularization term rather than to the shared adapter, the rank constraint or the training recipe — which is what the derivation predicts: with no penalty the protected

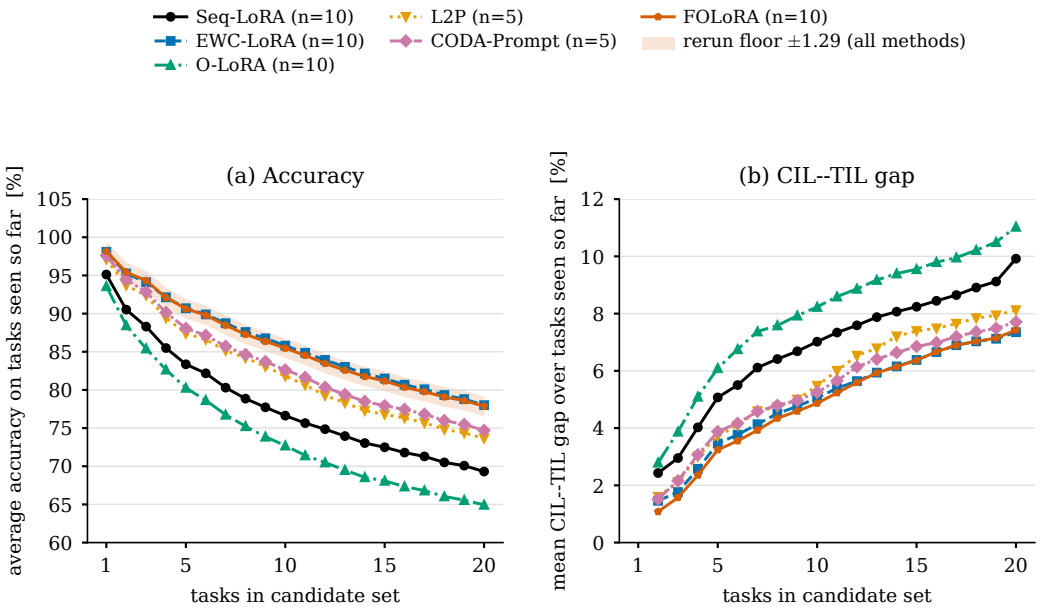


Figure 2: Average accuracy over the candidate classes seen so far (left) and the mean CIL–TIL gap over those candidate sets (right) on Split CIFAR-100. Every point is a single feature extraction of the *terminal* model re-scored against a growing candidate set; the horizontal axis is the number of tasks in that candidate set (1 to 20, i.e. 5 to 100 candidate classes at 5 classes per task), not training time, so the leftmost point already contains the forgetting incurred over the whole sequence. Each curve is drawn at the configuration *and* the seed budget of its row in Table 1 — ten seeds for Seq-LoRA, EWC-LoRA, O-LoRA and FOLoRA, five for L2P and CODA-Prompt — so its right-hand endpoint reproduces that row's $\overline{\text{ACC}}$. Six of the eight rows of Table 1 are plotted; InfLoRA and the SimpleCIL floor are tabulated but not drawn here, so the legend and the table do not cover an identical row set. Those budgets differ between methods (Sec. 5.1), so the legend carries each curve's n and the figure implies no equal n. No smoothing or interpolation is applied — the markers are the per-task measurements. The shaded band is the run-to-run reproducibility floor of this measurement (1.29 points, Sec. 5.4) — a property of the protocol rather than of any one method, and it applies to every curve shown. The vertical axis is bounded below at 60, i.e. 4.94 points under the lowest value plotted, rather than at zero; the full range is 60–105.

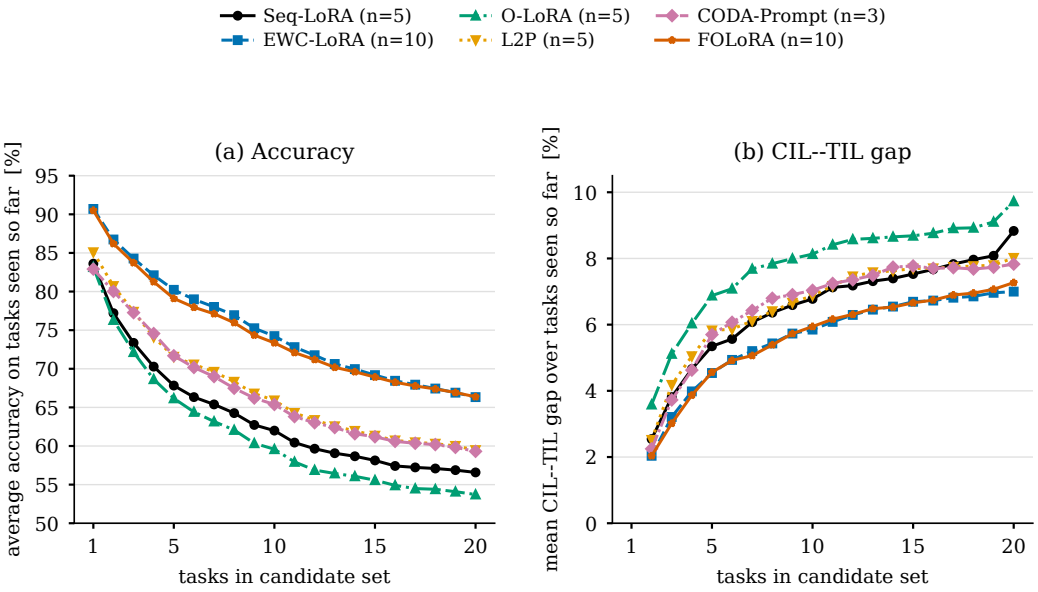


Figure 3: Average accuracy over the candidate classes seen so far (left) and the mean CIL–TIL gap over those candidate sets (right) on Split ImageNet-R. As in Fig. 2, each point is a single feature extraction of the terminal model re-scored against a growing candidate set — the number of tasks in it, 1 to 20, i.e. 10 to 200 candidate classes at 10 classes per task — not a training trajectory. Each curve is drawn at the configuration *and* the seed budget of its row in Table 1 — CODA-Prompt on three seeds, L2P, O-LoRA and Seq-LoRA on five, EWC-LoRA and FOLoRA on ten — so its right-hand endpoint reproduces that row's $\overline{\text{ACC}}$. As in Fig. 2, six of the eight rows of Table 1 are plotted; InfLoRA and the SimpleCIL floor are tabulated but not drawn. Those budgets differ between methods (Sec. 5.1), so the legend carries each curve's n and the figure implies no equal n. No reproducibility-floor band is drawn, because that floor was measured on CIFAR-100 only and we do not transfer it to another dataset. The vertical axis is bounded below at 50, i.e. 3.72 points under the lowest value plotted.

Table 3: Ablation on Split CIFAR-100. Δ and p compare each setting with the pre-registered main configuration (k=64, λ=3) using a paired t-test over common seeds; n is each row's own seed budget, so Δ is computed on the intersection and need not equal the difference of the two means shown. λ=0 removes the Fisher-weighted orthogonality penalty and changes nothing else. GAP is the CIL–TIL gap of Table 1, defined there; lower is better. The printed p is uncorrected; * marks $p < 0.05$ *after* Holm–Bonferroni correction over all CIFAR-100 configurations tested (Sec. 5.1), so a row may print $p < 0.05$ without a star.

Setting	n	$\overline{\text{ACC}}$	GAP	Δ vs main	p
λ *at* k=64					
λ=0	5	69.59±1.55	9.53±0.65	+8.72	0.0005*
λ=1	5	77.33±1.13	7.56±0.80	+0.99	0.1106
λ=3	10	77.87±0.90	7.42±0.59	*(main)*	—
λ=10	5	78.24±0.64	6.87±0.30	+0.08	0.8868
k *at* λ=3					
k=16	5	78.14±0.68	7.32±0.61	+0.17	0.7487
k=64	10	77.87±0.90	7.42±0.59	*(main)*	—
k=128	5	77.66±0.63	7.49±0.42	+0.65	0.2768

subspace is empty and the method degenerates to Seq-LoRA.

At k=64 the λ response is a step, not a peak, over the range we probed. Accuracy rises from 69.59 at $\lambda = 0$ to 77.33 at $\lambda = 1$ and 77.87 at $\lambda = 3$, and does not move beyond that: $\lambda = 10$ prints 78.24 on the five seeds it was run on, but its *paired* difference against the main configuration on those same five seeds is +0.08 ($p = 0.887$) — the two are indistinguishable, and the 0.37-point gap between the printed means is a seed-budget difference, not a gain. Only $\lambda = 0$ separates from the main configuration ($p = 0.0005$); $\lambda \in \{1, 10\}$ are each within noise of it ($p = 0.111$ and $p = 0.887$). We therefore do not claim an interior optimum at k=64 — the grid does not resolve one, and any ordering among $\lambda \in \{1, 3, 10\}$ is inside the run-to-run band of Sec. 5.4. The paired tests behind these accuracies are in Table 3.

At k=16 the decline at large λ is visible. Extending the sweep on the smaller protected subspace, where the penalty acts on fewer directions, average accuracy peaks at $\lambda \in \{3, 10\}$ (78.14 and 77.90, both n=5) and then declines before plateauing: 75.35 at $\lambda = 100$ (n=3), 73.61 at $\lambda = 300$ (n=10) and 73.85 at $\lambda = 1000$ (n=3) — the last two sit inside the run-to-run band, so we read the response as saturating rather than continuing to fall. This is the interior optimum that Remark 2 predicts, and it is the reason we treat the large-λ end as a real failure mode rather than as a harmless over-regularization. The seed budgets differ across these points, so we read the shape rather than the individual gaps; the paired tests behind them are in Table A.6.

k is a memory–accuracy dial, not a knob to tune. Accuracies at $k \in \{16, 64, 128\}$ differ by less than one point (78.14, 77.87, 77.66) and

no paired difference is significant ($p \geq 0.2768$, the k=128 comparison; Table 3), while the importance state grows linearly in k — 47.8 MiB at k=16 against 182.8 MiB at k=64 on the accounting convention of Table 4 (protected directions plus reference snapshot). We report k=64 because it was the pre-registered main configuration, not because it scores highest — it does not.

Fisher weighting versus equal weighting is not settled by our runs. Replacing the eigenvalue weights with equal weights per protected direction changes the picture with λ: at $\lambda = 3$ the Fisher-weighted variant is ahead by +3.58 points ($p = 0.019$) over three common seeds, while at $\lambda = 10$ the two are indistinguishable (-0.03, $p = 0.985$, on the same three seeds; note this is the Fisher-weighted versus equal-weight contrast, not the λ=10 row of Table 3, which compares that setting against the main configuration over five). With three seeds we claim no ordering in either direction; the comparison would need the main grid's seed budget before it could carry one, and we flag it here as an open question rather than leaving it out. Both pairings are listed in Table A.6, at the common-seed count rather than at either arm's full budget.

5.4. Limitations and discussion

Reproducibility floor. We measured the run-to-run spread directly rather than describing it qualitatively. Two runs on CIFAR-100 share the same seed (0) and the same configuration field by field — their `config.json` files agree on all 21 fields except the `tag`, and the one other field that differs (`folora_lambda`) is not read by the EWC method at all, which we confirmed in the source — so any difference between them is non-determinism

rather than a hyperparameter effect. The gap is 77.26 versus 75.97: 1.29 **points at the same seed and the same configuration**. That number is not an outlier: the 10-seed spread of the same configuration is ±1.08, so it sits inside the distribution. The consequence governs how every comparison in this paper must be read: *the effects we measure on CIFAR-100 are smaller than this single-run noise band*, so a single seed, or any subset of seeds chosen after the fact, cannot support a claim in either direction. This is why we report paired t-tests over seeds throughout and why we describe any difference inside the band as parity rather than as a win — including differences that happen to favor us.

Where the variance comes from. On CIFAR-100 the accuracy of FOLoRA is consistent in mean but noisier across seeds than the regularized baselines it is compared against — ±0.90 for FOLoRA against ±0.54 for EWC-LoRA and ±0.64 for InfLoRA (Table 1) — while being *less* noisy than unprotected sequential training (±1.92 for Seq-LoRA and ±3.03 for O-LoRA at its initialization-only setting), which is consistent with the penalty damping seed-to-seed drift rather than amplifying it. We attribute the residual variance to two sources. First, the importance kernel is estimated from a finite per-sample budget (up to 300 gradients per task), and its top-k approximation is sensitive to the particular samples that land in the tails of the estimate. Second, under a frozen backbone, a fixed-rank (r=16) adapter and no replay, the class ordering exerts a strong effect on the final subspace, so individual seeds drift apart. We further found that the epoch budget interacts with the method family: adapter-based Fisher methods degrade under longer training because over-fitting drives the per-sample gradients toward

zero and thereby degenerates the estimated Fisher, whereas prompt-based methods benefit from longer training of their prompt keys. This motivates the per-family epoch budgets in Sec. 5.1. A natural direction for future work is a stochastic (Monte-Carlo) Fisher estimate and class-order averaging to tighten the variance.

What we claim, in two layers. Because the CIFAR effect sizes sit inside the noise band above, we separate the claims into one layer the evidence supports unconditionally and one it supports only with an explicit caveat.

The first layer is about *what the method does*, and does not depend on beating anyone. (i) Removing the Fisher-weighted penalty and changing nothing else returns the method to the level of unprotected sequential fine-tuning, so the entire effect of FOLoRA is carried by the regularization term it adds rather than by the shared adapter or the training recipe (Sec. 5.3). (ii) The protected rank is not a sensitive knob: accuracies at $k \in \{16, 64, 128\}$ differ by less than one point and none of the differences is significant, while the importance state grows linearly in k — which makes k a memory–accuracy dial rather than a hyperparameter to tune. (iii) On CIFAR-100 FOLoRA reaches accuracy *on par with the strongest regularized baseline* rather than above it; we state parity as parity. Against EWC-LoRA, the comparison flips sign depending on which of the tuned EWC operating points is taken as the baseline, so we report all of them in Table 1 and let the reader see the flip rather than selecting the point that reads best.

The second layer is where FOLoRA is *ahead*, and it holds against the parameter-efficient family rather than against full-parameter regularization: against L2P, CODA-Prompt, InfLoRA and O-LoRA, the paired differences

are significant on both benchmarks and seed-wise all of one sign. That claim is the one we treat as the paper's positive result; it is also the one where the baseline's seed budget is smallest, so Sec. 5.1 records the seed sets explicitly for each family instead of implying a single shared one.

Cost: trainable parameters are traded for importance state. FOLoRA's efficiency claim is narrower than "memory-efficient", and we quantify the trade-off rather than leaving it implicit. Its *trainable* parameter count is constant in the number of tasks — the same adapter is reused and re-projected — and the low-rank update merges back into the frozen backbone, so inference costs nothing extra. Against that, the importance model is a per-layer rank-k subspace rather than EWC's per-parameter diagonal, and it is larger. Table 4 states both sides with a single accounting convention: the importance column counts *everything a method must keep resident to compute its penalty*, i.e. the Fisher plus the parameter anchor for EWC, and the accumulated directions plus the reference snapshot for FOLoRA. On that convention FOLoRA at k=64 is 32× EWC's importance state; at k=16 it is 8.5×. This is the price of representing direction correlations, which a diagonal preconditioner cannot do at any rank, and it is why we report k as a dial rather than claiming a memory advantage.

Scope. Three limitations bound the generality of everything above. First, the evidence covers two benchmarks and a single backbone (ViT-B/16); we make no claim about whether the curvature-weighted subspace transfers to convolutional backbones or to domain-incremental shifts, where the class-incremental protocol we evaluate under does not directly apply. Second, the baselines are reproduced under one shared protocol and one shared backbone

Table 4: Resource accounting. Trainable parameters are counted at the first and at the last task (T=1 and T=20). These totals include the classifier head, which grows with the number of classes for *every* row — that growth is a property of the class-incremental protocol and is shared by all methods, so the column that separates them is "new module per task", not the totals. "Importance" is the state a method must keep resident to compute its penalty, counted on one convention for all rows — the diagonal Fisher *plus* the parameter anchor for EWC-LoRA, and the accumulated protected directions *plus* the reference snapshot for FOLoRA (on the same byte counts, EWC-LoRA's Fisher alone is 2.8 MiB, and FOLoRA's directions alone are 45.0 MiB at k=16 and 180.0 MiB at k=64). L2P and CODA-Prompt build their prompt pool once and keep it fixed as tasks arrive, so they too add no module per task: like the LoRA-based rows, their growth from T=1 to T=20 is the shared classifier head.

Method	Trainable params		New module	Importance
	T=1	T=20	per task	state
Seq-LoRA	741,125	814,180	no	—
EWC-LoRA	741,125	814,180	no	5.6 MiB
O-LoRA	741,125	14,822,500	yes	—
InfLoRA	741,125	14,822,500	yes	—
L2P	115,205	188,260	no	—
CODA-Prompt	99,845	172,900	no	—
FOLoRA (k=16)	741,125	814,180	no	47.8 MiB
FOLoRA (k=64)	741,125	814,180	no	182.8 MiB

rather than at their published settings, which is what makes the comparison internally fair but also means the numbers are not the numbers reported in the original papers; where a baseline's own configuration is materially different — the prompt methods' pool size and prompt length — we state both configurations, so that the comparison cannot be read as a deliberately weakened baseline. Training budgets are set per family rather than per method (Sec. 5.1), so the adapter-based baselines that Table 1 compares against FOLoRA share its budget; we report no baseline row at a budget we chose to disadvantage it. One gap in that protocol is worth stating explicitly rather than leaving for a referee to find: the five-epoch adapter budget is justified by the behaviour of the *Fisher-weighted* methods, yet it is applied to the non-Fisher adapter baselines as well, and we do not report Seq-LoRA, O-LoRA or InfLoRA at a longer budget — so whether they would improve at twenty epochs is not settled by these experiments. We flag it as an open protocol question rather than claiming the asymmetry cannot matter. with one exception, and we report that exception and its resolution rather than a summary of the family. At inference we merge every per-task adapter for both O-LoRA and InfLoRA, which is the setting their papers specify. On ImageNet-R that places both above the floor (53.72 for O-LoRA and 58.93 for InfLoRA, against 49.71); on CIFAR-100 it places InfLoRA above it (72.95 against 70.31) while the tabulated O-LoRA row remains below it (64.94). That row implements O-LoRA's orthogonal *initialization* without the training-time objective the original paper specifies — the formulation augments the loss with $\lambda_1 \sum_{i<t} \|A_t^\top A_i\|_F^2$ — so we ran that objective at two constraint strengths, λ_1=0.1 and λ_1=1.0 (n=3 each). It moves the CIFAR-

100 mean from 64.94 to 70.84 ± 0.60 and 72.54 ± 0.43, and reaches 56.30 and 61.15 on ImageNet-R: above the floor on both benchmarks. Against these stronger configurations FOLoRA's advantage narrows from +12.92 to +7.53 and +5.83 on CIFAR-100, and from +13.20 to +10.59 and +5.74 on ImageNet-R, and it remains significant at both strengths after correction within the two-member family these variants form — so we report both strengths for each benchmark rather than the one that suits us best, and the narrower figures are the margin we are entitled to claim. Both baselines are evaluated at the per-family budget Sec. 5.1 fixes, so no row of Table 1 is reported at a budget chosen to disadvantage it. What separates these configurations from FOLoRA is not the family but what the constraint acts on: an orthogonality penalty placed in the adapter input space bounds $A_t^\top A_i$ and leaves the merged update $\sum_i B_i A_i$ unconstrained, whereas FOLoRA's Fisher-weighted term penalizes the directions in which that update moves.

The bound's slack. Finally, Sec. 4 reports that the forgetting bound is not tight at the operating point we probe: its remainder and its stationarity residual are the same order as its leading term, and the finite-sample Fisher estimate is not a calibrated surrogate for the measured curvature. The derivation is therefore used as a statement of *which* quantity to penalize, and the empirical support for the method is the ablation above rather than the bound. We prefer to state this explicitly, since the alternative — presenting a loose bound as if it were predictive — is the failure mode that a reader who rederives Eq. (15) would find anyway.

6. Conclusion

We propose FOLoRA, a replay-free method for continual learning that mitigates catastrophic forgetting in low-rank adaptation while keeping the number of trainable parameters fixed as the number of tasks grows. FOLoRA maintains an accumulated second-order importance kernel for each adapted layer, forms an importance-weighted protected subspace from its top eigendirections, and penalizes alignment of the adapter's displacement with those directions in proportion to their importance. We derive a forgetting upper bound that decomposes into Fisher-weighted subspace overlaps and showed that FOLoRA's regularizer minimizes its dominant term, thereby providing a principled stability–plasticity trade-off and containing equal-weight orthogonalization as a special case. We also measure the slack of that bound instead of assuming it away: at the operating point we probe the remainder and the stationarity residual are the same order as the quadratic term, so the bound is presented as a design principle for what to penalize rather than as a tight or numerically predictive characterization of forgetting. The empirical case for the regularizer rests on the λ=0 ablation, which removes the Fisher-weighted penalty while changing nothing else and returns the method to the level of unprotected sequential fine-tuning. Experiments on standard class-incremental benchmarks show that FOLoRA reaches an average accuracy on par with the strongest regularized baseline while using no replay buffer and no growth in the parameter count, and we quantify the importance-model memory it pays for representing that model as a rank-k subspace rather than EWC's diagonal one — 32× EWC's importance state at k=64 and 8.5× at k=16, which is the trade-off that makes k a memory–accuracy dial rather

than a free parameter. We also report the run-to-run reproducibility floor we measured (1.29 points at a fixed seed and a fixed configuration), and we state the resulting conclusions at the strength the data supports: parity with the strongest regularized baseline, rather than an improvement over it, and a significant advantage over the parameter-efficient prompt and subspace baselines.

A natural direction for future work is to extend FOLoRA to task-agnostic settings with distribution shift (domain-incremental learning) and to language models, where the low-rank subspace geometry of different task families can be exploited more explicitly.

CRediT authorship contribution statement

Jiang Tao: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study (CIFAR-100 and ImageNet-R) are publicly available standard continual-learning benchmarks. The source code implementing the proposed method, all baselines, the evaluation protocol and the scripts that reproduce every table and figure in this paper are openly

available at https://github.com/ml-universe/FOLoRA. The repository ships the code, the configuration files, and the evaluation outputs under reports/ from which the result tables are generated, so every reported number can be re-derived without retraining; it does not redistribute the benchmark datasets or the trained checkpoints, which are large and are regenerated by the training scripts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to improve the language and readability of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendix A. Supporting statistics

Sec. 5.1 fixes the confirmatory family and states the correction policy. This appendix lists the individual comparisons behind that family, so that none of the verdicts has to be taken on trust or reconstructed from prose. Table A.5 *is* the confirmatory family: FOLoRA against each baseline of Table 1, on average accuracy, with the uncorrected and the Holm-corrected p side by side. Table A.6 collects the comparisons the body quotes in prose but no other table covers — the same-λ Fisher-weighting contrast, the k=16 sweep, and the two orthogonality strengths of O-LoRA — either because they are not against the main configuration or because their seed budgets differ from it. The two are kept apart rather than merged into one table because their correction conventions differ. Table A.5 is the confirmatory family and reports corrected p-values throughout. Table A.6 is exploratory: it is quoted uncorrected, with the single exception its caption states — the two O-LoRA orthogonality points form a two-member family per benchmark that the body quotes *with* its within-family correction, so that block's p_{Holm} column is filled in and the remaining rows leave it blank. A corrected p from one family should not be read against a corrected p from another.

Three conventions hold throughout. n is always the number of seeds *common* to the two arms of that row, never a shared budget, so rows with different n do not have the same power and their p-values should not be compared as if they did. Δ is FOLoRA minus the comparison, in accuracy points, so positive favours FOLoRA; on the gap metric reported in the body the sign convention is the opposite, which is one reason the two are tabulated separately. And a row whose n is below two carries no test at all: it is listed

with p given as n/a rather than dropped, because the reason a point cannot support a claim is part of the record.

Appendix B. The probe behind Sec. 4.4

Sec. 4.4 states the conclusion of the bound-tightness probe and the main text uses only that conclusion. This appendix carries the protocol and the individual measurements, so that the claim can be checked rather than taken on trust.

Protocol. All numbers below come from an in-training probe that, at each task boundary of a 20-task run, fits the expansion of the accumulated past-task loss along the *realized* adapter displacement, $\Delta\mathcal{L}(\alpha\,\delta\theta) = a\alpha + b\alpha^2 + c\alpha^3 + \mathcal{O}(\alpha^4)$, over $\alpha \in [0, 1.5]$. Measuring the nonlinearity along the realized displacement, rather than bounding L_3, is what makes the remainder checkable at all: L_3 is not recoverable from any stored checkpoint, whereas a, b and c are. We report a and c as ratios to $|b|$, so that "is this correction small?" is read directly off the number. The probe run is FOLoRA at λ=10, k=16, seed 0 on Split CIFAR-100, giving 19 usable boundaries. It is a single configuration, and we do not have the seed spread needed to say how much of what follows is run-to-run variation; the numbers should be read as one operating point, not as a distribution over the method.

(i) The remainder is not negligible. The ratio $|c|/|b|$ has median 0.163, minimum 0.010 and maximum 2.46, with 5 of the 19 boundaries above 0.30. Under the decision rule frozen before these numbers were computed (median ≤ 0.10: keep the negligibility claim; $0.10 <$ median ≤ 0.30: weaken it;

Table A.5: The confirmatory family of Sec. 5.1: FOLoRA against each baseline of Table 1 on average accuracy, with both the uncorrected p and the Holm–Bonferroni corrected p_{Holm} that the paper's verdicts rest on. Conventions are stated above; the exploratory comparisons are in Table A.6.

Benchmark	Comparison	n	Δ	p	p_{Holm}
Confirmatory family: Table 1, average accuracy					
CIFAR-100	Seq-LoRA	10	+8.56	0.0000	0.0000
CIFAR-100	EWC-LoRA (lam=100)	10	+0.65	0.0734	0.2934
CIFAR-100	EWC-LoRA (lam=300)	10	-0.14	0.6440	0.9921
CIFAR-100	EWC-LoRA (lam=1000)	10	+0.21	0.4961	0.9921
CIFAR-100	EWC-LoRA (lam=3000)	10	+0.47	0.2507	0.7521
CIFAR-100	O-LoRA	10	+12.92	0.0000	0.0000
CIFAR-100	L2P	5	+4.67	0.0002	0.0013
CIFAR-100	CODA-Prompt	5	+3.61	0.0016	0.0080
CIFAR-100	InfLoRA	5	+5.37	0.0002	0.0014
ImageNet-R	Seq-LoRA	5	+10.34	0.0002	0.0011
ImageNet-R	EWC-LoRA (lam=100)	10	+2.85	0.0002	0.0011
ImageNet-R	EWC-LoRA (lam=300)	10	+0.05	0.9010	1.0000
ImageNet-R	EWC-LoRA (lam=1000)	10	+0.04	0.9235	1.0000
ImageNet-R	O-LoRA	5	+13.20	0.0000	0.0000
ImageNet-R	L2P	5	+7.45	0.0000	0.0000
ImageNet-R	CODA-Prompt	3	+7.58	0.0022	0.0066
ImageNet-R	InfLoRA	3	+7.96	0.0011	0.0046

Table A.6: The remaining comparisons the body quotes in prose, which no other table covers: variants of the main configuration and of one baseline, on both benchmarks. These are *exploratory* — they were chosen after seeing the results — so the p given here is uncorrected and the family each belongs to is stated where its number is used in Sec. 5.2. The one exception is the O-LoRA training-time orthogonality block, whose two λ_1 points form a two-member family that the body quotes *with* its within-family correction, so its p_{Holm} column is filled in. The columns are those of Table A.5, and the same conventions apply. No entry here is a re-analysis: the same seeds and the same cached features are used throughout.

Benchmark	Comparison	n	Δ	p	p_{Holm}
Exploratory (uncorrected p)					
Fisher-weighted against equal-weighted, same λ					
CIFAR-100	λ=3	3	+3.58	0.0190	—
CIFAR-100	λ=10	3	-0.03	0.9852	—
k=16 protected directions against the k=64 main configuration					
CIFAR-100	λ=3	5	+0.17	0.7487	—
CIFAR-100	λ=10	5	+0.41	0.3832	—
CIFAR-100	λ=30	1	+0.36	n/a	—
CIFAR-100	λ=100	3	+3.02	0.1489	—
CIFAR-100	λ=300	10	+4.25	0.0000	—
CIFAR-100	λ=1000	3	+4.52	0.0561	—
O-LoRA orthogonality strength (p_{Holm}: two-member family)					
CIFAR-100	λ_1=0.1	3	+7.53	0.0018	0.0035
CIFAR-100	λ_1=1	3	+5.83	0.0129	0.0129
ImageNet-R	λ_1=0.1	3	+10.59	0.0172	0.0344
ImageNet-R	λ_1=1	3	+5.74	0.0174	0.0344

> 0.30: drop it), this lands in the middle band, and the honest reading is the one the frozen rule prescribes: the cubic term is *of the same order as, but does not dominate*, the quadratic term. The tail is the part that matters — at the worst boundary the quantity the bound calls a remainder is $2.5\times$ the quantity FOLoRA minimizes.

(ii) The stationarity residual is not small either. The ratio $|a|/|b|$ has median 0.99 and maximum 23.1. Assumption 1 is stated for the *unregularized* loss, so at a regularized endpoint the residual is $-\lambda\nabla\Omega$, and Remark 2 argued that it should vanish proportionally to the displacement. It does not vanish fast enough to be dropped: at the realized displacement the linear term is comparable to the quadratic one, so the "quadratic plus corrections" reading of the expansion is not numerically available at this λ.

(iii) The Fisher estimate is not a calibrated surrogate for the curvature. The ratio between the term the bound prescribes, $\frac{1}{2}\delta\theta^\top \bar{F}\delta\theta$, and the quadratic coefficient b fitted from the same measurement has median 2.8×10^{-5}: the finite-sample, rank-k, block-diagonal estimate of F_τ underestimates the measured curvature of the past loss along the realized displacement by four to five orders of magnitude. It is worth being precise about what this does and does not say. It does *not* contradict Theorem 1, which is stated for the exact F_τ; it says that Assumption 3 is far from innocuous at this operating point, which is precisely the regime the empirical-Fisher literature warns about [71]. It is also, in part, the intended behaviour seen from the other side: the displacement is close to orthogonal to the top-k Fisher subspace *by construction*, since suppressing that alignment is what the penalty does, so the Fisher-weighted quadratic form is small along it. Both readings agree on

the operative point: the bound's leading term is not a quantitative predictor of forgetting in this setting.